CHAPTER 11

Biomolecules

Carbohydrates

1. Sucrose on hydrolysis gives: *(2020)*

a. α-D-Glucose + β-D-Glucose

b. α-D-Glucose + β-D-Fructose

c. α-D-Fructose + β-D-Fructose

d. β-D-Glucose + α-D-Fructose

2. Which of the following statement is not true about glucose? *(2020-Covid)*

a. It contains five hydroxyl groups

b. It is a reducing sugar

c. It is an aldopentose

d. It is an aldohexose

3. The difference between amylose and amylopectin is *(2018)*

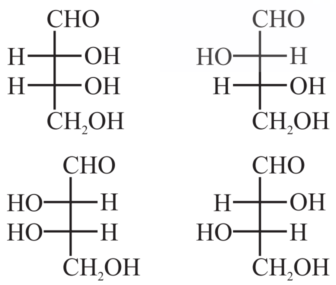
a. Amylopectin have 1 → 4 α-linkage and 1 → 6 α-linkage

b. Amylose have 1 → 4 α-linkage and 1 → 6 β-linkage

c. Amylose is made up of glucose and galactose

d. Amylopectin have 1 → 4 α-linkage and 1 → 6 β-linkage

4. The correct corresponding order of names of four aldoses with configuration given below: *(2016 - II)*



a. L-erythrose, L-threose, L-erythrose, D-threose

b. D-threose, D-erythrose, L-threose, L-erythrose

c. L-erythrose, L-threose, D-erythrose, D-threose

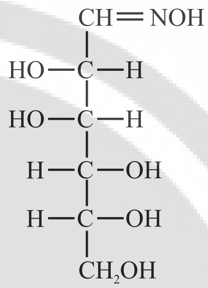
d. D-erythrose, D-threose, L-erythrose, L-threose

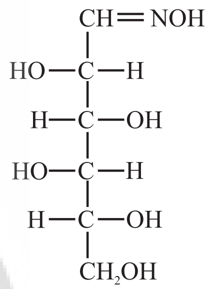
5. Which one given below is a non-reducing sugar? *(2016 - I)*

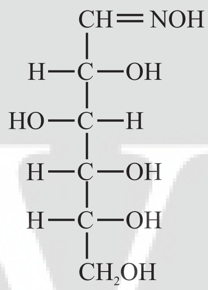
a. Sucrose b. Maltose

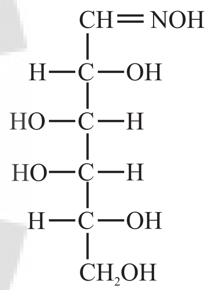
c. Lactose d. Glucose

6. D(+) glucose reacts with hydroxyl amine and yields an oxime. The structure of the oxime would be: *(2014)*

a. 

b. 

c. 

d. 

Proteins

7. Which of the following is a basic amino acid? *(2020)*

a. Alanine b. Tyrosine

c. Lysine d. Serine

8. The non-essential amino acid among the following is: *(2019)*

a. Valine b. Leucine

c. Alanine d. Lysine

9. Which of the following compounds can form a zwitter ion? *(2018)*

a. Aniline b. Acetanilide

c. Glycine d. Benzoic acid

10. In a protein molecule, various amino acids are linked together by: *(2016 - I)*

a. Dative bond

b. α-glycosidic bond

c. β-glycosidic bond

d. Peptide bond

Enzymes and Vitamins

11. The **incorrect** statement regarding enzymes is *(2022)*

a. Enzymes are very specific for a particular reaction and substrate

b. Enzymes are biocatalysts

c. Like chemical catalysts enzymes reduce the activation energy of bio process.

d. Enzymes are polyssaccharides.

12. The RBC deficiency is deficiency disease of: *(2021)*

a. Vitamin B_6 b. Vitamin B_1

c. Vitamin B_2 d. Vitamin B_{12}

13. Deficiency of which vitamin causes osteomalacia? *(2020-Covid)*

a. Vitamin D b. Vitamin K

c. Vitamin E d. Vitamin A

Nucleic Acids

14. The central dogma of molecular genetics states that the genetic information flows from: *(2016 - II)*

a. Amino acids → Proteins → DNA

b. DNA → Carbohydrates → Proteins

c. DNA → RNA → Proteins

d. DNA → RNA → Carbohydrates

15. The correct statement regarding RNA and DNA, respectively is: *(2016 - I)*

a. The sugar component in RNA is 2′-deoxyribose and the sugar component in DNA is arabinose

b. The sugar component in RNA is arabinose and the sugar component in DNA is 2′-deoxyribose

c. The sugar component in RNA is ribose and the sugar component in DNA is 2′-deoxyribose

d. The sugar component in RNA is arabinose and the sugar component in DNA is ribose

Hormones

16. Which of the following hormones is produced under the conditions of stress which stimulate glycogenolysis in the liver of human beings? *(2014)*

a. Thyroxin b. Insulin

c. Adrenaline d. Estradiol

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
b	c	a	d	a	c	c	c	c	d	d	d	a	c	c	c